Week 2

Computer Platform

Understanding the foundation of digital systems

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What Is Inside Your Computer?

Look beyond the screen: computing depends on coordinated physical parts.

DISCUSSION

- **Q1** What makes the device work?
- **Q2** Is hardware alone enough?
- **Q3** What connects applications to hardware?



Are All of These Computers?

A computer platform can be visible, mobile, remote, embedded, or hidden.

Smartphone

Classify: computer, platform, or both?

Laptop

Classify: computer, platform, or both?

Smartwatch

Classify: computer, platform, or both?

Server

Classify: computer, platform, or both?

Smart TV

Classify: computer, platform, or both?

Car controller

Classify: computer, platform, or both?

VOTE FIRST • EXPLAIN LATER

From Digital Technology to Computer Platform

Most digital services depend on a platform underneath.



Key message

A platform is the environment that enables software to run, interact with hardware, and provide useful digital services.

Learning Outcomes

By the end of Week 2, students should be able to explain and apply these ideas.

✓ **LO1**

Define computer platform

✓ **LO2**

Identify hardware components

✓ **LO3**

Explain CPU, memory, storage, and I/O

✓ **LO4**

Distinguish CPU, GPU, and AI accelerators

✓ **LO5**

Describe client–server and cloud previews

✓ **LO6**

Select a platform for a simple scenario

What is a Computer Platform?

A platform is more than a machine.

Definition

A computer platform is the combination of hardware architecture and software environment that allows applications to operate.

Hardware

Physical computing resources

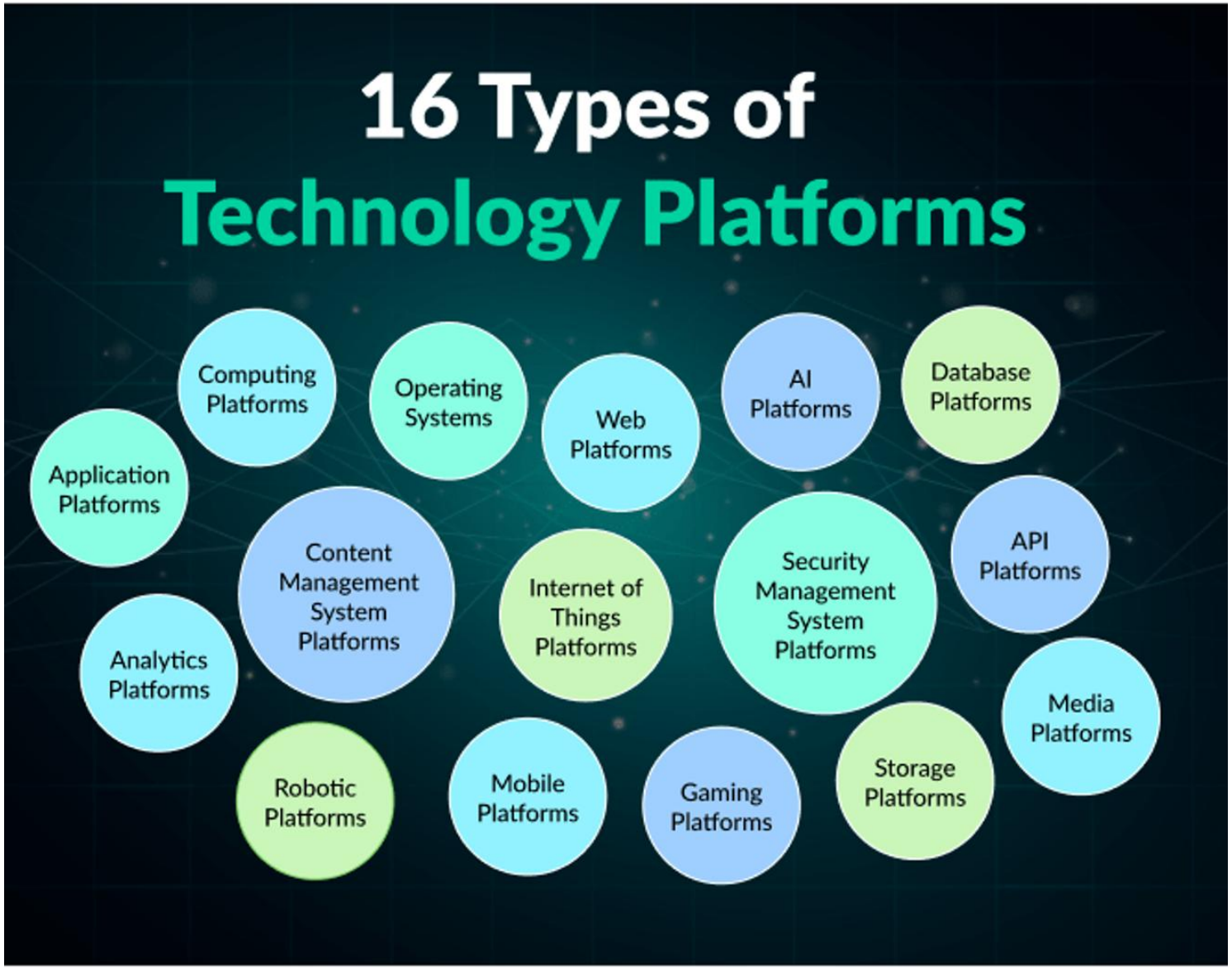
System software

OS, drivers, libraries, runtime

Applications

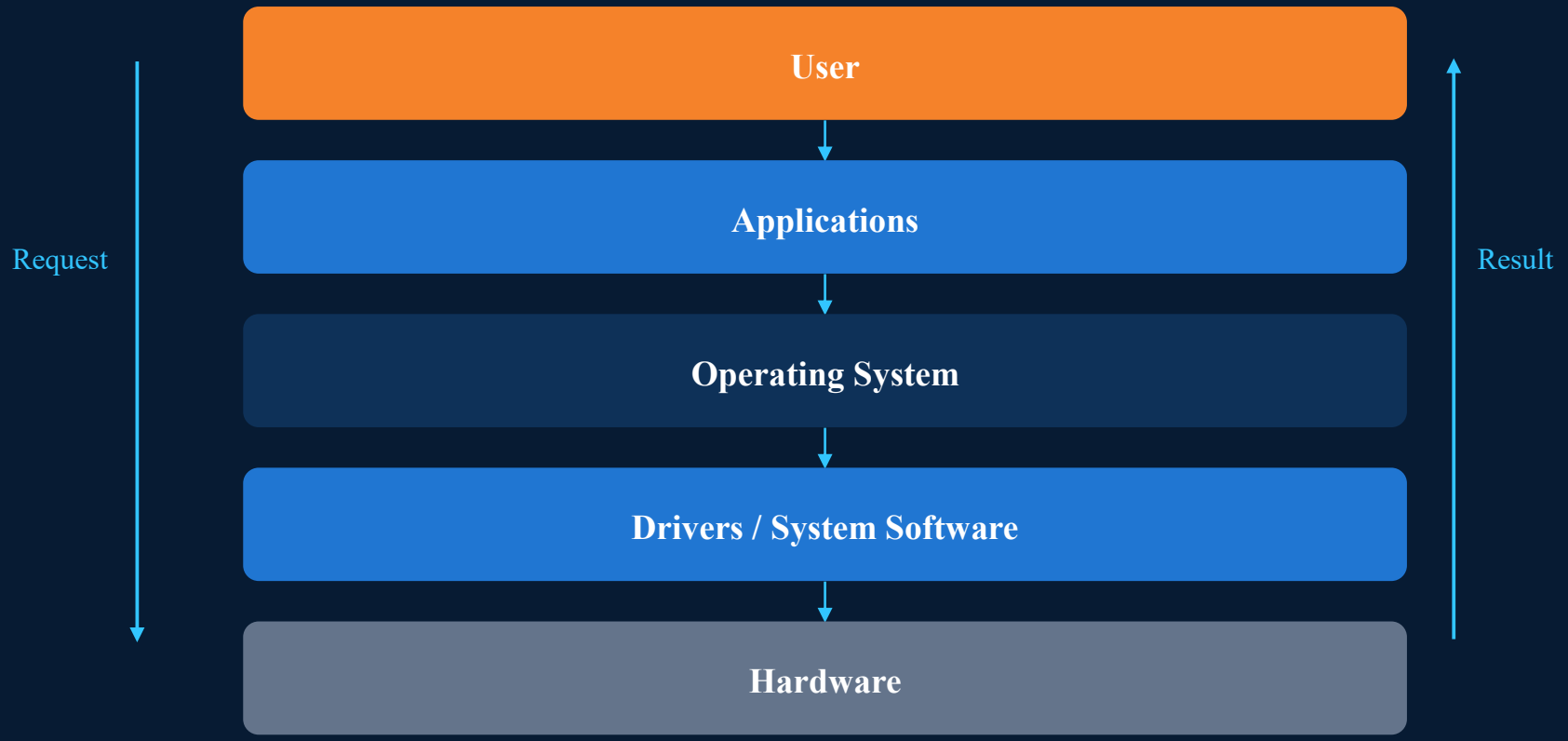
Programs that users actually use

PLATFORM = HARDWARE + SOFTWARE ENVIRONMENT


16 Types of
Technology Platforms
Computing Platforms
Operating Systems
Web Platforms
AI Platforms
Database Platforms
Application Platforms
Content Management System Platforms
Internet of Things Platforms
Security Management System Platforms
API Platforms
Analytics Platforms
Media Platforms
Robotic Platforms
Mobile Platforms
Gaming Platforms
Storage Platforms

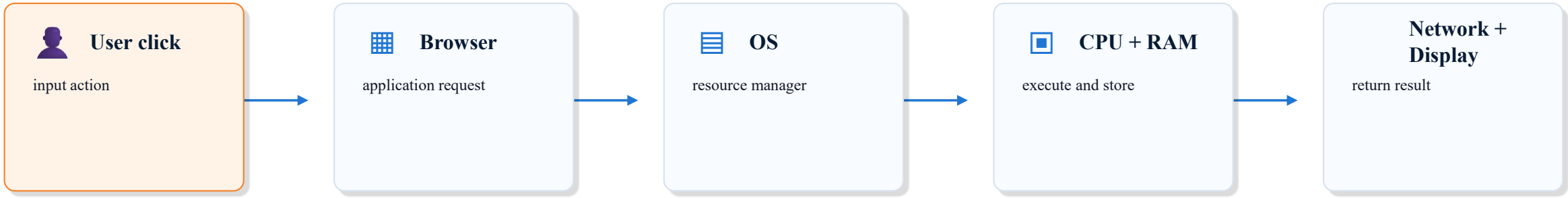
Platform as a Stack

Requests flow downward; results flow back upward.



How the Layers Work Together

Example: opening a web browser.



Takeaway

Applications feel simple to the user because the platform hides complex coordination among hardware and software.

Platform ≠ Device

The same physical form can support different platform environments.

Physical device

The visible object: laptop, phone, desktop, server, or embedded controller.

Computing platform

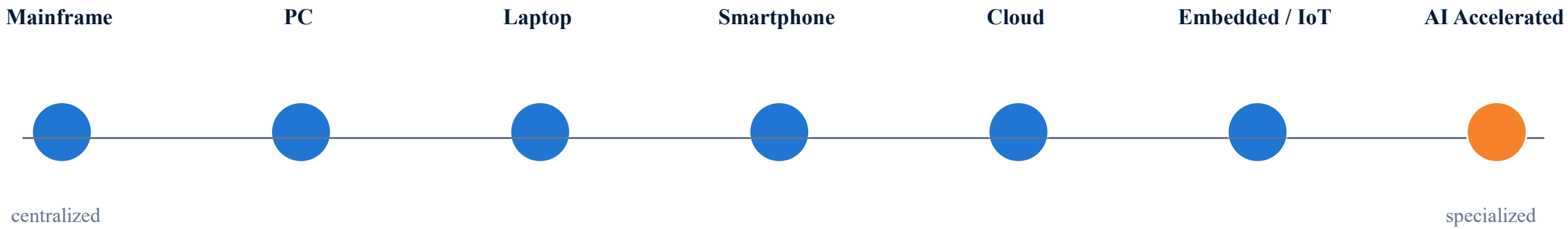
Device hardware plus OS, drivers, runtimes, APIs, and applications.

Example

Laptop hardware + operating system + drivers + software frameworks = usable platform.

Computer Platform Evolution

Computing moved from one visible machine to distributed, specialized ecosystems.



⟷ Context

Today’s platform may span your device, remote servers, embedded controllers, and specialized chips.



02

Hardware & Organization

How physical parts become a coordinated computer system

Hardware: Inside and Outside

Every platform combines internal components with human-facing devices.

Internal hardware

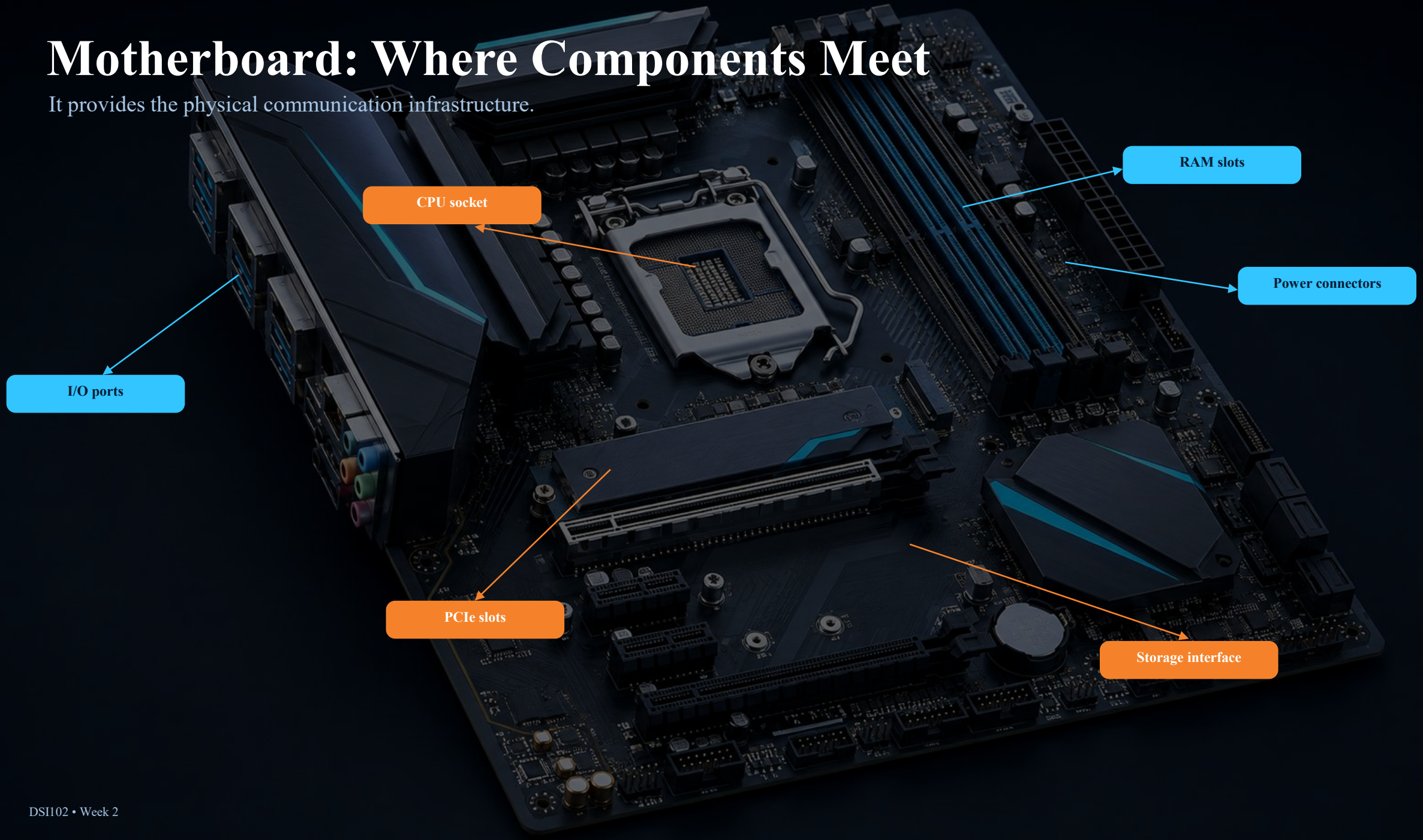
CPU
GPU / accelerators
RAM
Storage
Motherboard
Power
Communication interfaces

External hardware

Keyboard
Mouse / touch
Display
Camera
Microphone
Printer
External storage

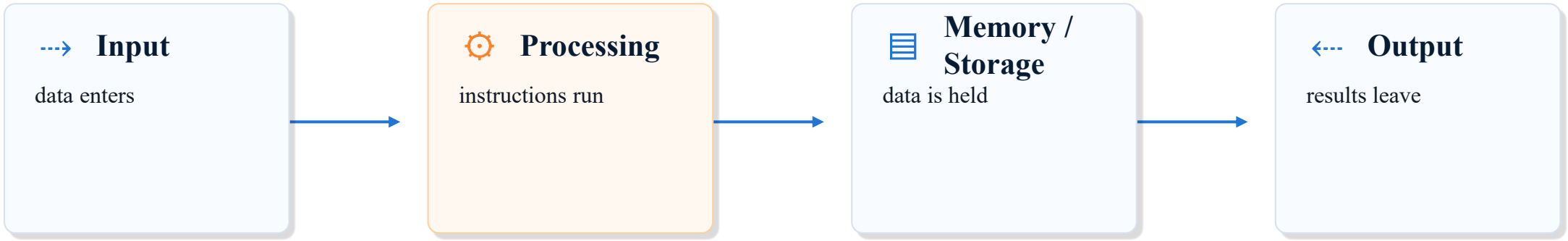
Motherboard: Where Components Meet

It provides the physical communication infrastructure.



Basic Computer Organization

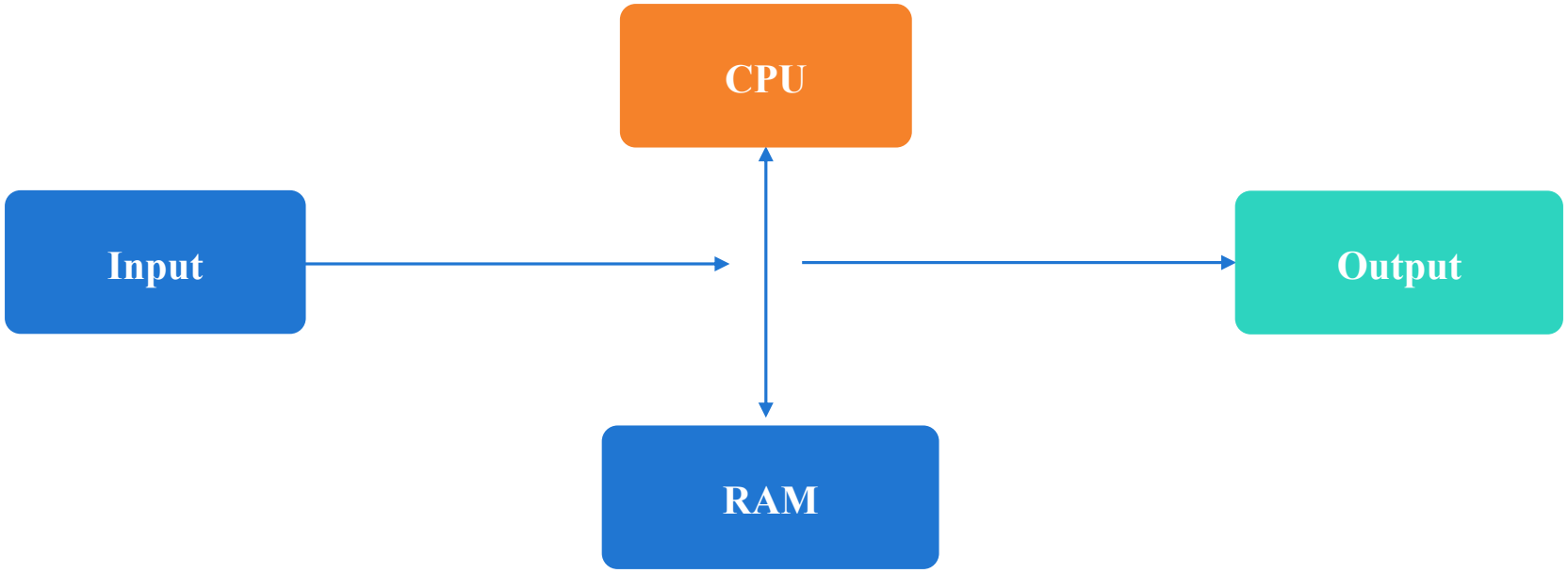
At a high level, computers transform input into useful output.



Input → Processing → Memory/Storage → Output

A Computer Is a System

No component works independently.



Architecture view

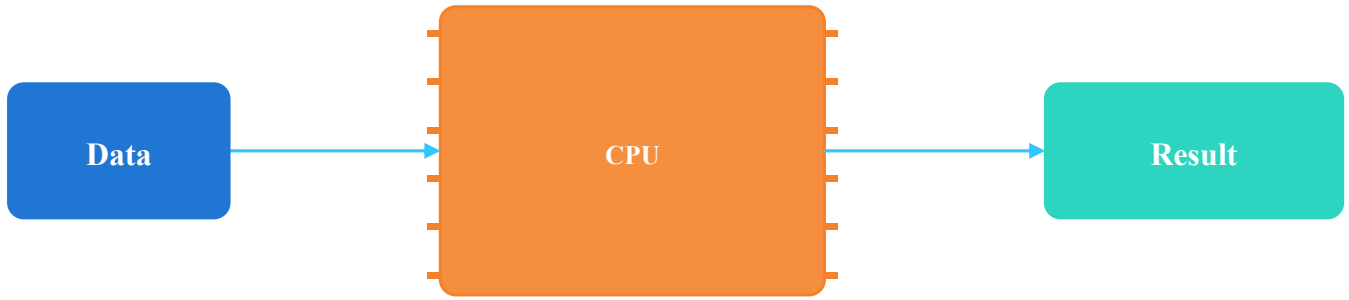
Components exchange data through pathways, interfaces, and control signals.

CPU: General-Purpose Processor

The CPU executes instructions and coordinates the system.

What it does

Executes program instructions
Coordinates system operations
Handles varied workloads
Works closely with memory



Main Parts of a CPU

A simplified model for introductory understanding.

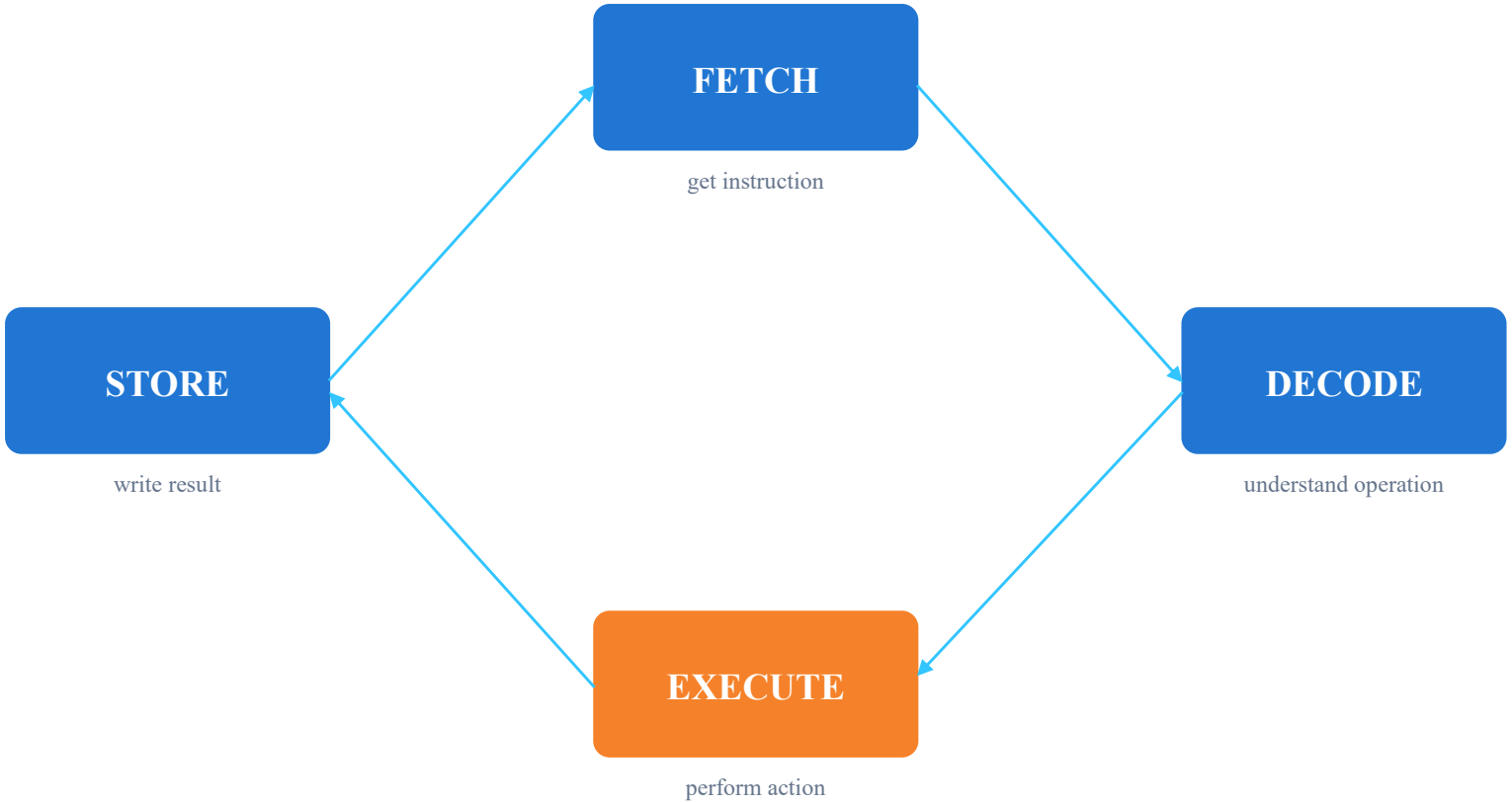
! Remember

CPU details matter, but Week 2 focuses on how the CPU fits into the platform.



Instruction Cycle

The CPU repeatedly fetches, decodes, executes, and stores.

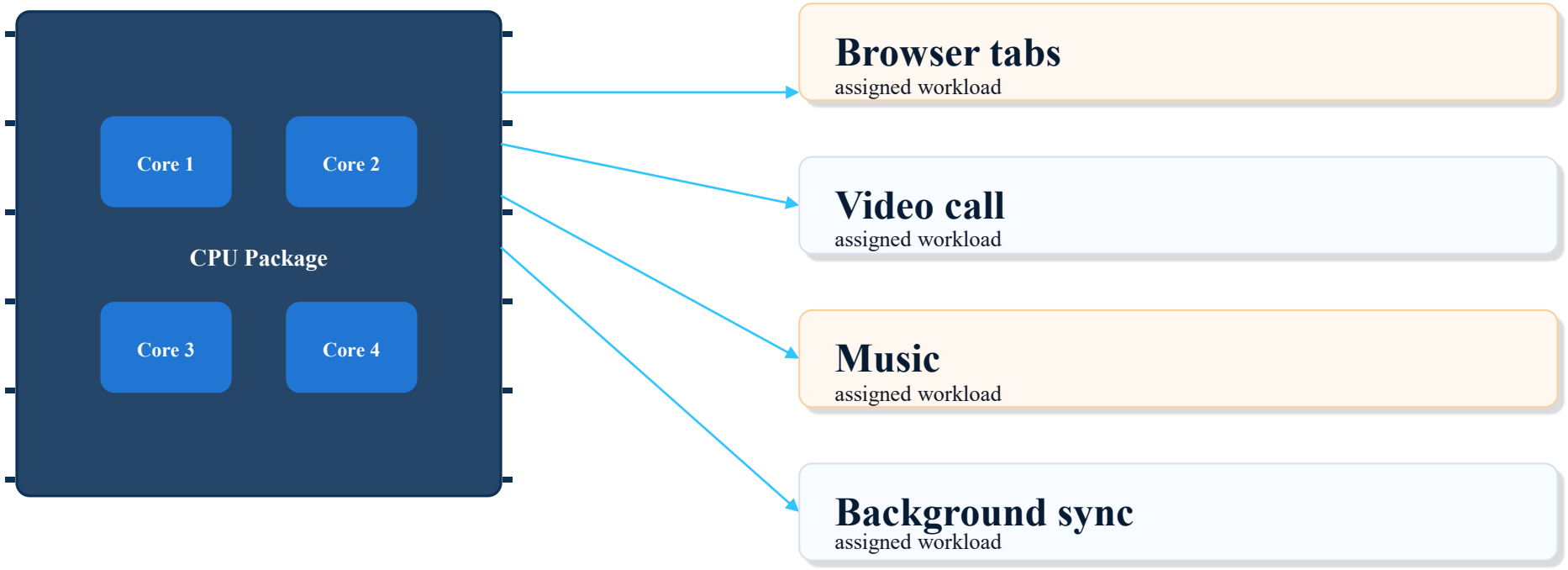


Simple example

For 2 + 3, the CPU fetches the instruction, decodes addition, executes it, then stores the result.

Multicore Computing

More cores help when work can be divided.



MY TH

Misconception check

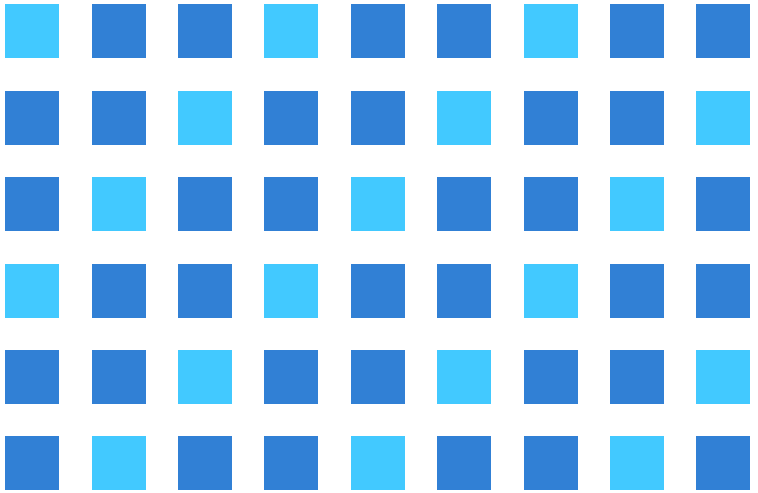
More cores do not automatically make every task faster. Software must be able to use parallel work.

GPU: Parallel Processing Engine

A GPU is designed for many similar calculations at once.

Common workloads

Graphics rendering
Video processing
3D design and games
AI computation
Scientific simulation



Visual idea

Many smaller processing units can handle large parallel workloads efficiently.

Specialized Processors for AI

TPU, NPU, and similar accelerators target specific computation patterns.

CPU

Flexible general work

GPU

Large parallel work

AI Accelerator

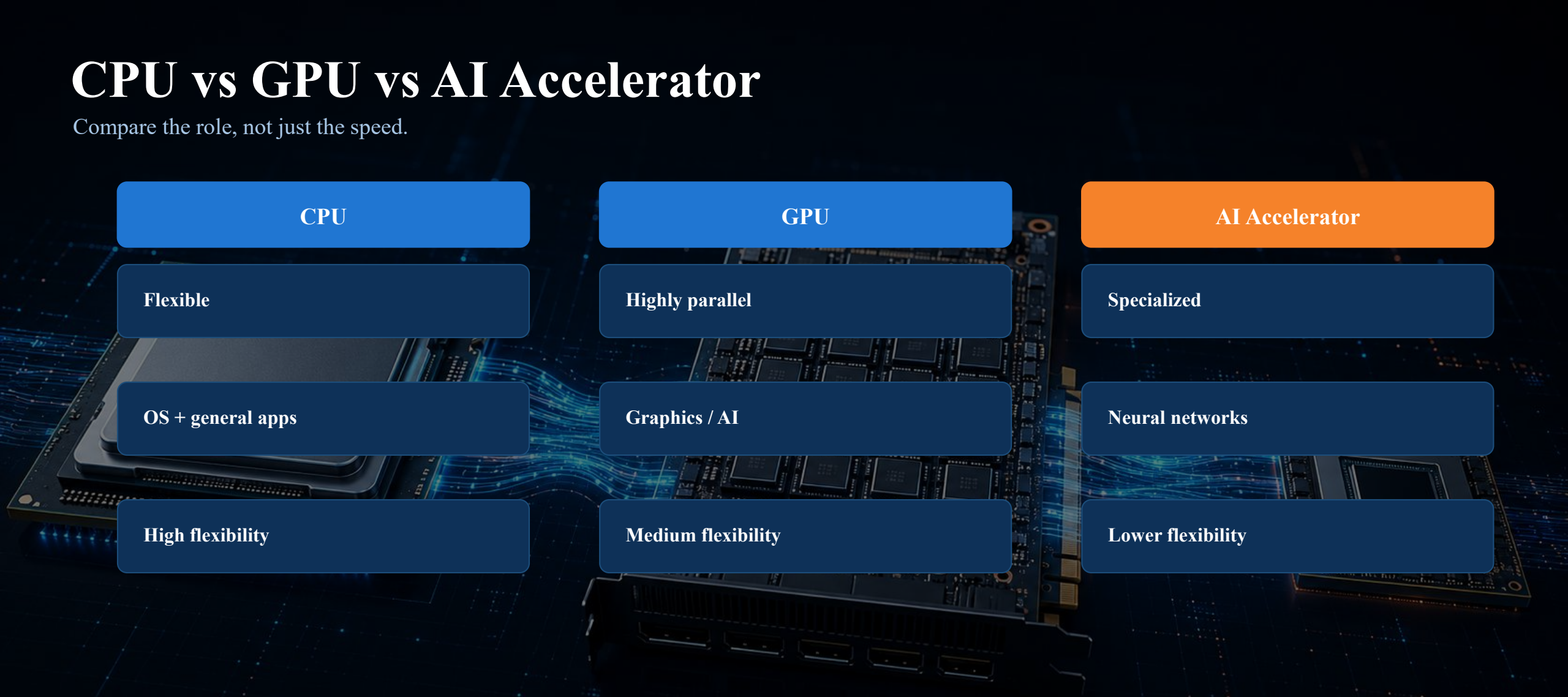
Matrix / tensor workloads

Key message

Different processors are optimized for different kinds of work. One is not universally “best.”

CPU vs GPU vs AI Accelerator

Compare the role, not just the speed.



CPU	GPU	AI Accelerator
Flexible	Highly parallel	Specialized
OS + general apps	Graphics / AI	Neural networks
High flexibility	Medium flexibility	Lower flexibility

One Device, Many Processors

Modern platforms combine general, parallel, and specialized chips.





03

Memory & Storage

Temporary work, persistent files, and the hierarchy between them

Memory vs Storage

The names sound similar, but the roles are different.

RAM = working table

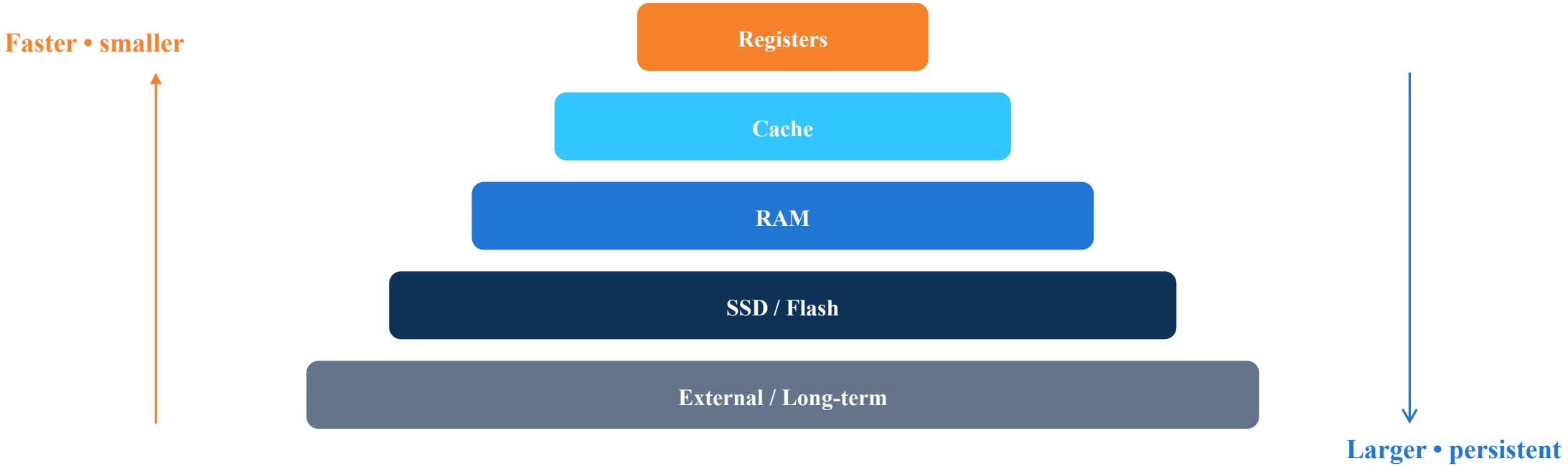
Temporary
Fast
Holds active data
Cleared when power is off

Storage = filing cabinet

Persistent
Usually larger
Holds files and apps
Keeps data without power

Memory Hierarchy

Closer to the CPU usually means faster, smaller, and more expensive per unit.



RAM: Temporary Workspace

More RAM can help when active work does not fit comfortably.

RAM properties

Temporary workspace
Volatile: loses data when power is removed
Holds running apps and active data

Small desk

few active items

Large desk

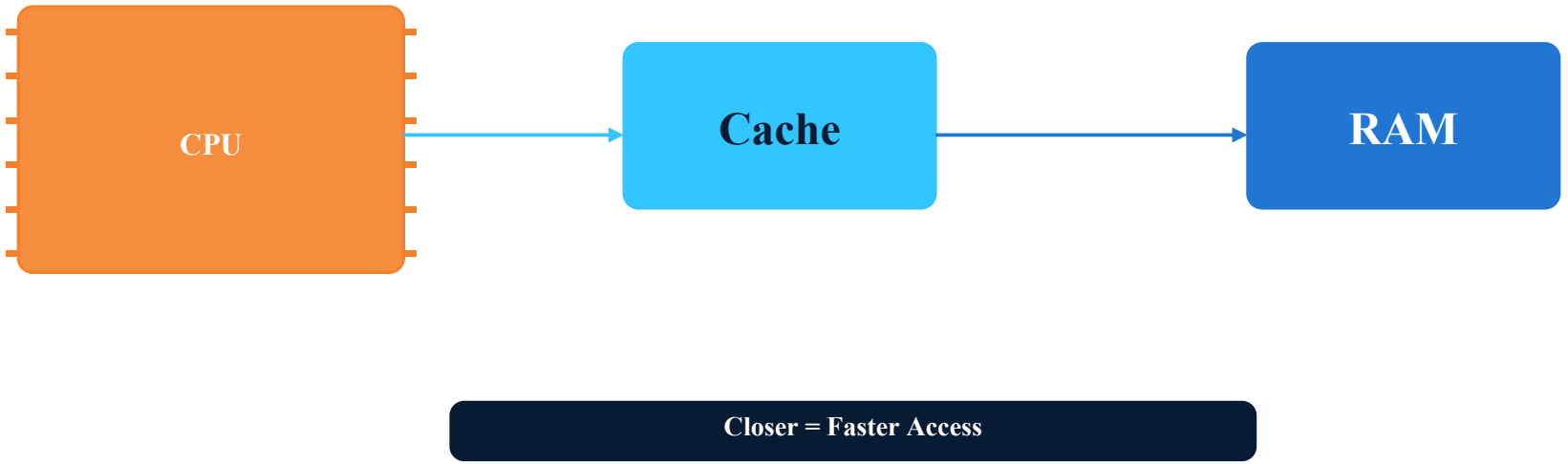
more active items

MYTH **Misconception**

More RAM does not make the CPU faster; it can reduce memory bottlenecks.

Cache Memory

Cache keeps frequently needed data close to the CPU.



↔ Concept

The CPU can process quickly only when data arrives quickly enough.

SRAM vs DRAM

Keep the comparison conceptual, not circuit-level.

S SRAM

Faster
More expensive
Lower density
Often used for cache

D DRAM

Slower than SRAM
Lower cost per bit
Higher density
Used for main memory

Place in hierarchy: CPU → Cache/SRAM → RAM/DRAM → Storage

Volatile vs Non-Volatile Memory

Computers need both temporary speed and persistent availability.

Volatile

Data disappears when power is removed.

Examples: registers, cache, RAM.

Non-Volatile

Data remains after power is removed.

Examples: firmware storage, flash, SSD.

ROM / Firmware idea

Some basic instructions must remain available before the operating system loads.

Storage: Persistent Data

Storage keeps applications, files, and system data after power is off.

HDD

Magnetic disk + moving parts

SSD

Flash memory, no moving disk

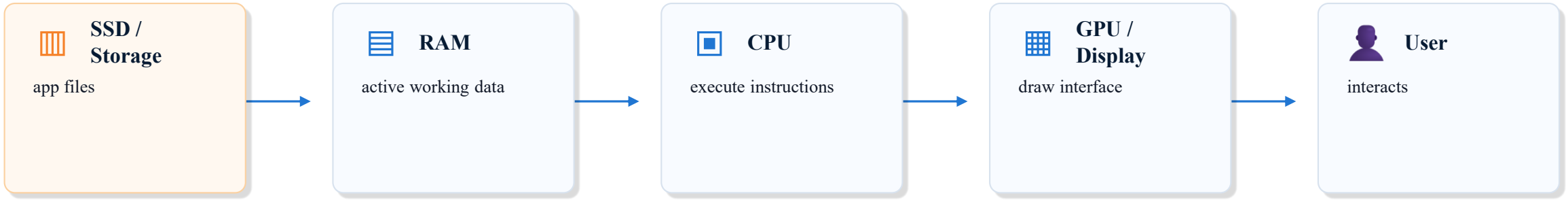
External / Cloud

Removable or remote persistence

Storage is persistent; RAM is temporary.

What Happens When You Open an App?

A visible click triggers an invisible data journey.



Platform view

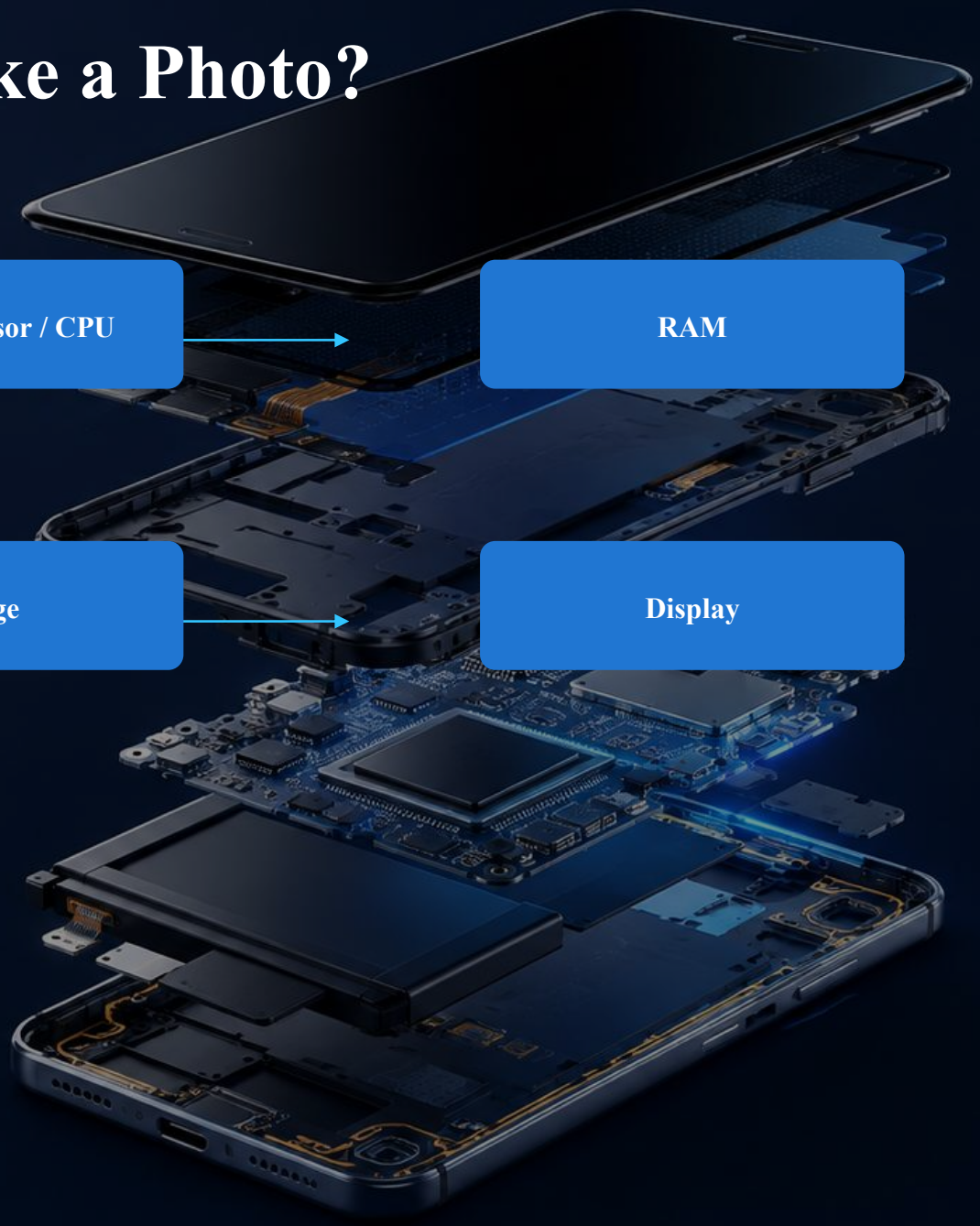
Opening an app uses storage, memory, processing, output hardware, and software coordination.

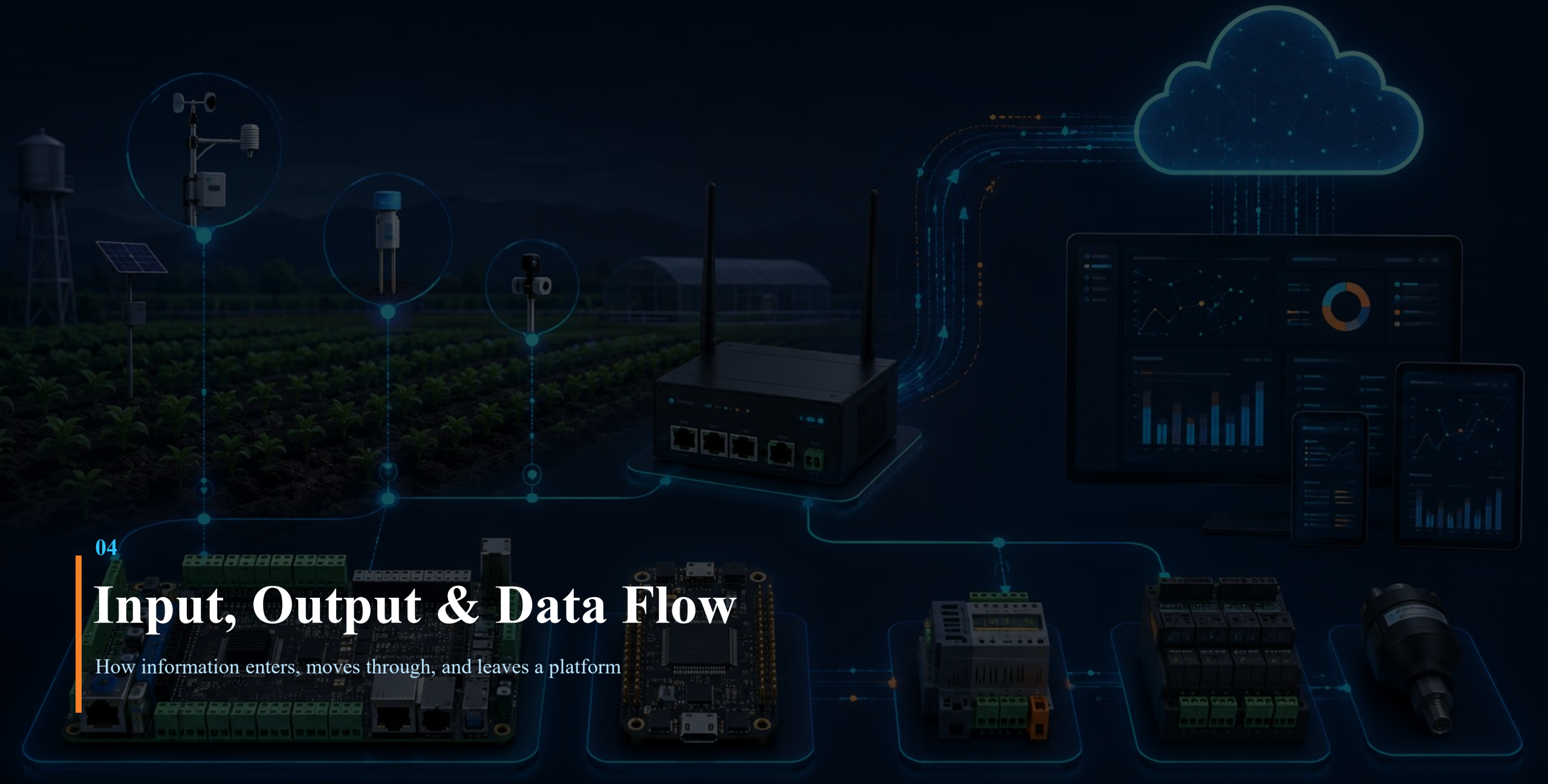
What Happens When You Take a Photo?

End-to-end synthesis: input, processing, memory, storage, and output.

Camera sensor → **Image processor / CPU** → **RAM**

AI / GPU enhancement → **Storage** → **Display**





04

Input, Output & Data Flow

How information enters, moves through, and leaves a platform

Input Devices: Data Enters the Platform

Input converts actions or physical signals into digital data.

Human commands

Keyboard • mouse • touchscreen

Audio

Microphone

Images / video

Camera • scanner

Machine-readable

Barcode • QR reader

Environment

Sensors

Example
Mouse evolution: mechanical → optical → laser. The interaction feels similar, but sensing technology changes.

Output Devices: Digital Data Becomes Human Experience

Output translates processed data into something people can see, hear, or feel.

Visual

Monitor • projector • VR/AR display

Audio

Speaker • headphones

Physical

Printer • haptic feedback

Clarify
A GPU or sound card processes output signals; the display or speaker is the human-facing output device.

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Interfaces Connect Components

A platform needs standard ways for parts to communicate.

USB

HDMI

PCIe

Bluetooth

Wi-Fi

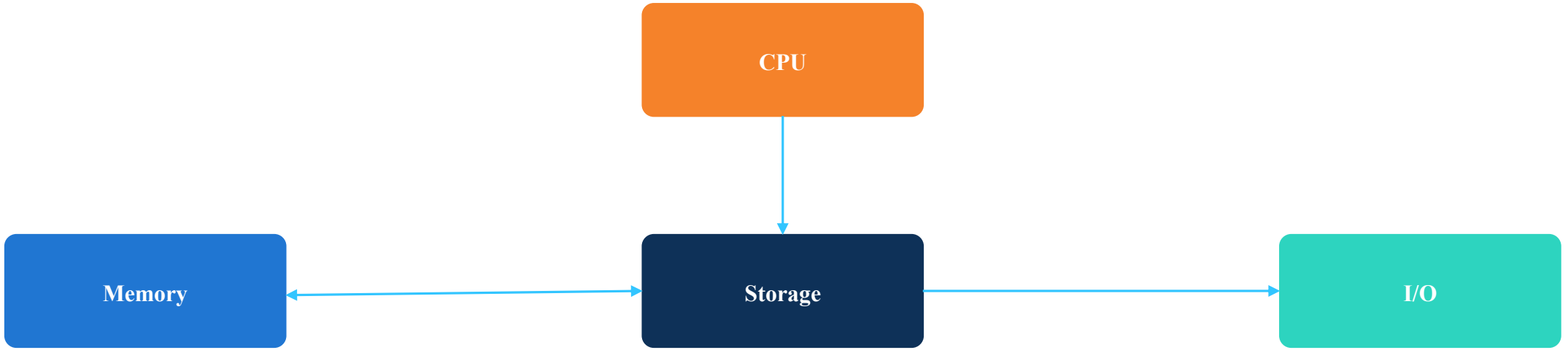
Ethernet

↔ Concept

Interfaces define how information moves between devices, components, and networks.

System Bus Concept

Pathways carry data, addresses, and control signals.



Data

actual information

Address

where data belongs

Control

when and how to move



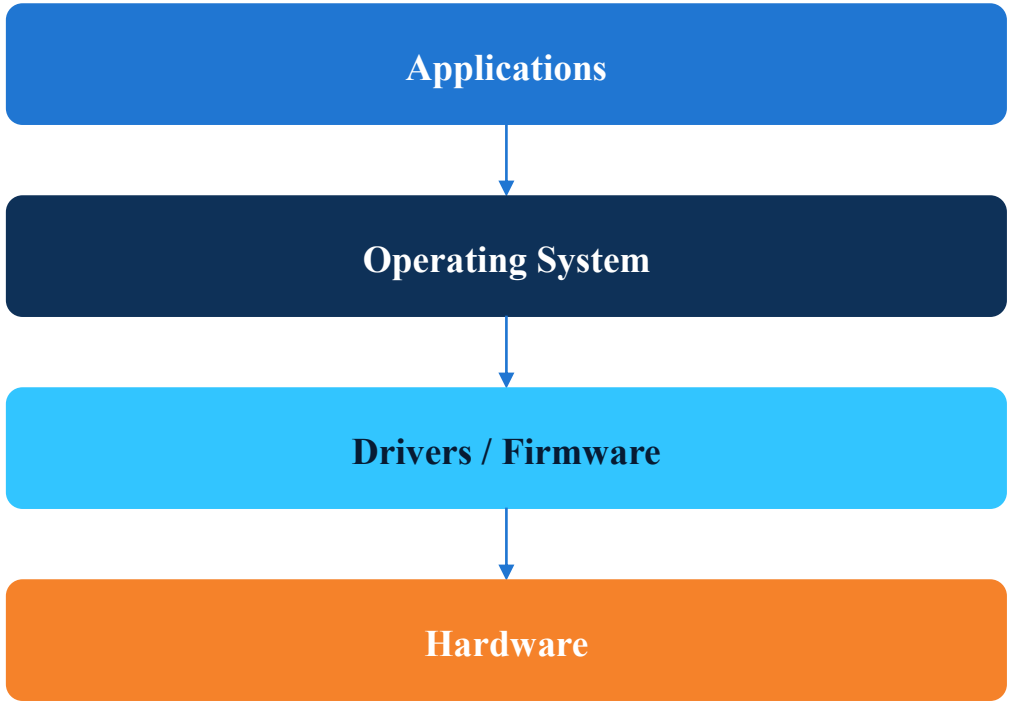
05

Software Turns Hardware into a Platform

Operating systems, applications, and services make the machine usable

Hardware Needs Software

Physical components are not useful until software can manage them.



Key message
The software stack exposes hardware capabilities to applications and users.

Operating System Preview

The OS is the manager between applications and hardware.

Manage CPU

Detailed OS concepts come next week.

Manage memory

Detailed OS concepts come next week.

Manage files

Detailed OS concepts come next week.

Manage devices

Detailed OS concepts come next week.

NEXT WEEK → Operating System

Application Types Depend on Platform

Where an app runs changes the platform it relies on.

Desktop app

Runs mainly on local PC

Mobile app

Runs on phone hardware + mobile OS

Web app

Runs in browser + server

Cloud app

Runs across remote infrastructure

Client–Server Platform

The platform may span multiple computers.



Example

Opening a university website uses your browser as a client and a remote machine as a server.

Servers Are Defined by the Service They Provide

A server is usually a role, not just a box.

Web server

delivers web content

Database server

stores structured data

File server

shares files

Mail server

handles email

Application server

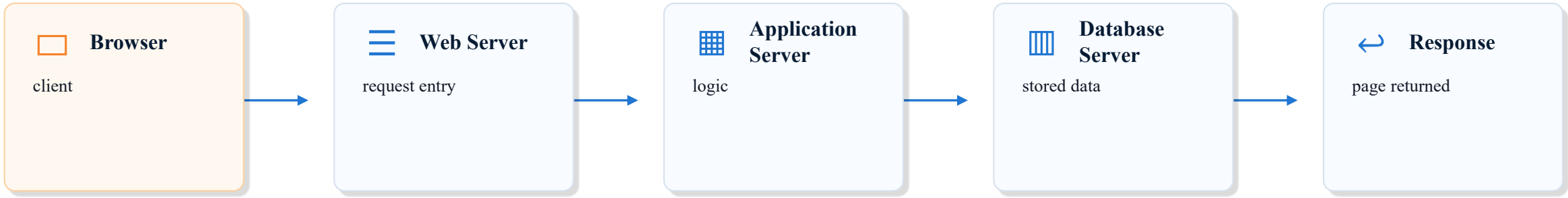
runs business logic

VM Virtual server

acts like separate machine

Example: Logging In to an Online Store

One action may involve several server roles.



Platform view
The platform for one service can include client hardware, network access, server software, and storage.



06

Platform Types

PC, mobile, server, embedded, IoT, and cloud platforms optimize for different needs

Personal vs Mobile Platforms

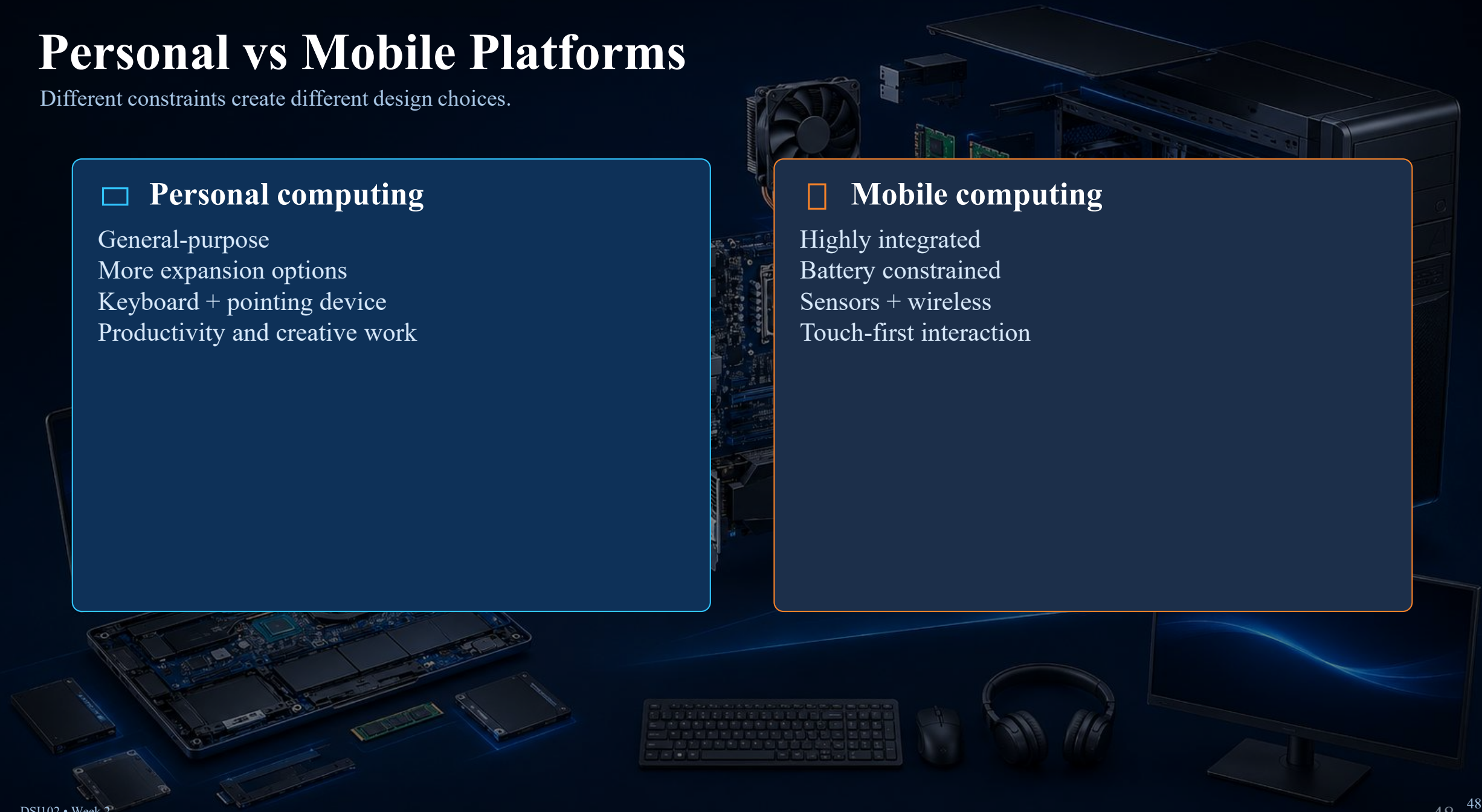
Different constraints create different design choices.

Personal computing

General-purpose
More expansion options
Keyboard + pointing device
Productivity and creative work

Mobile computing

Highly integrated
Battery constrained
Sensors + wireless
Touch-first interaction



Server Platforms

Built for many users, high workloads, and continuous operation.

- **Reliability**
- **Scalability**
- **Large memory / storage**
- **Network access**
- **Continuous service**



Embedded & IoT Platforms

Many computers are hidden inside other products.

Embedded system

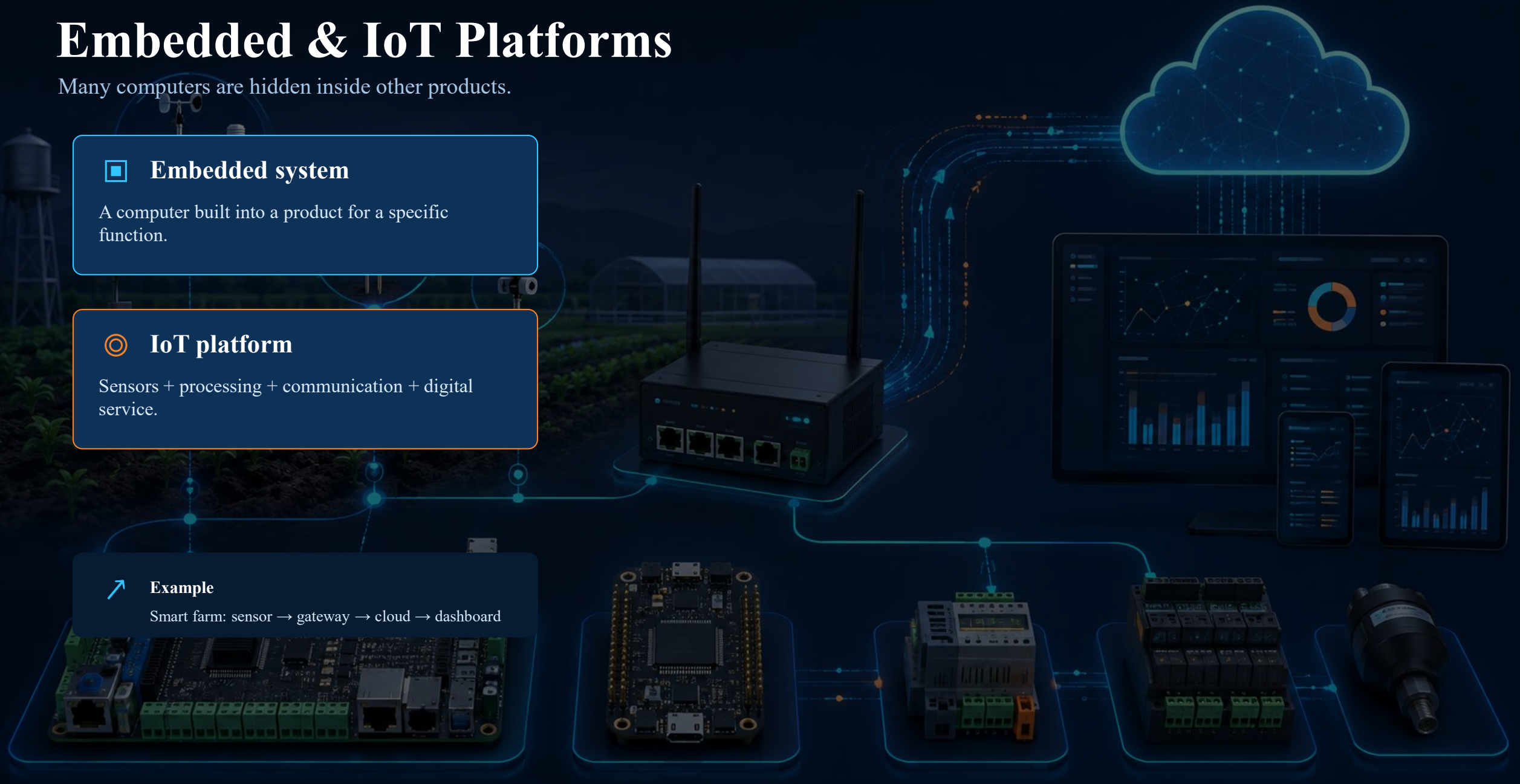
A computer built into a product for a specific function.

IoT platform

Sensors + processing + communication + digital service.

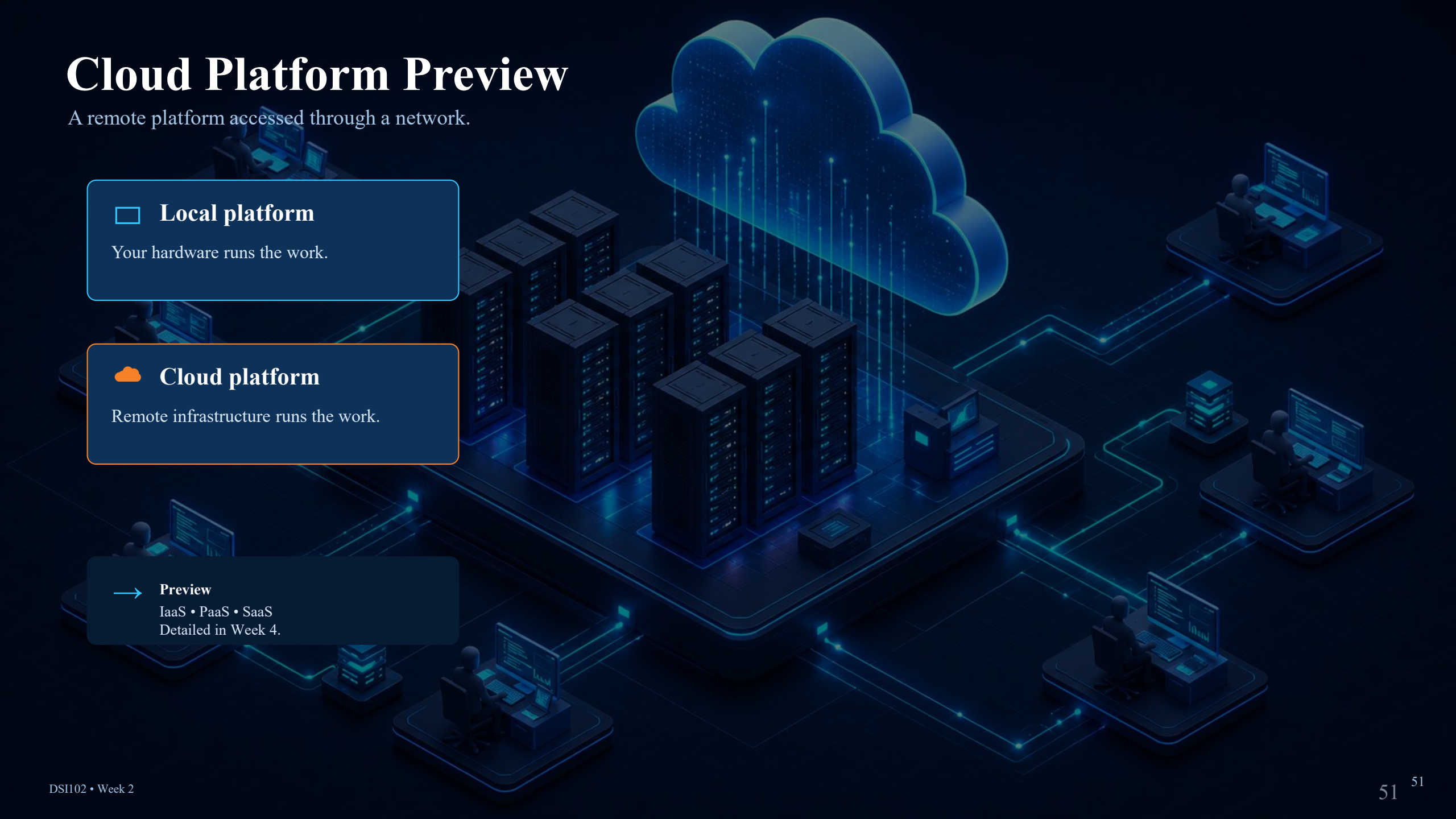
Example

Smart farm: sensor → gateway → cloud → dashboard



Cloud Platform Preview

A remote platform accessed through a network.



Local platform

Your hardware runs the work.

Cloud platform

Remote infrastructure runs the work.

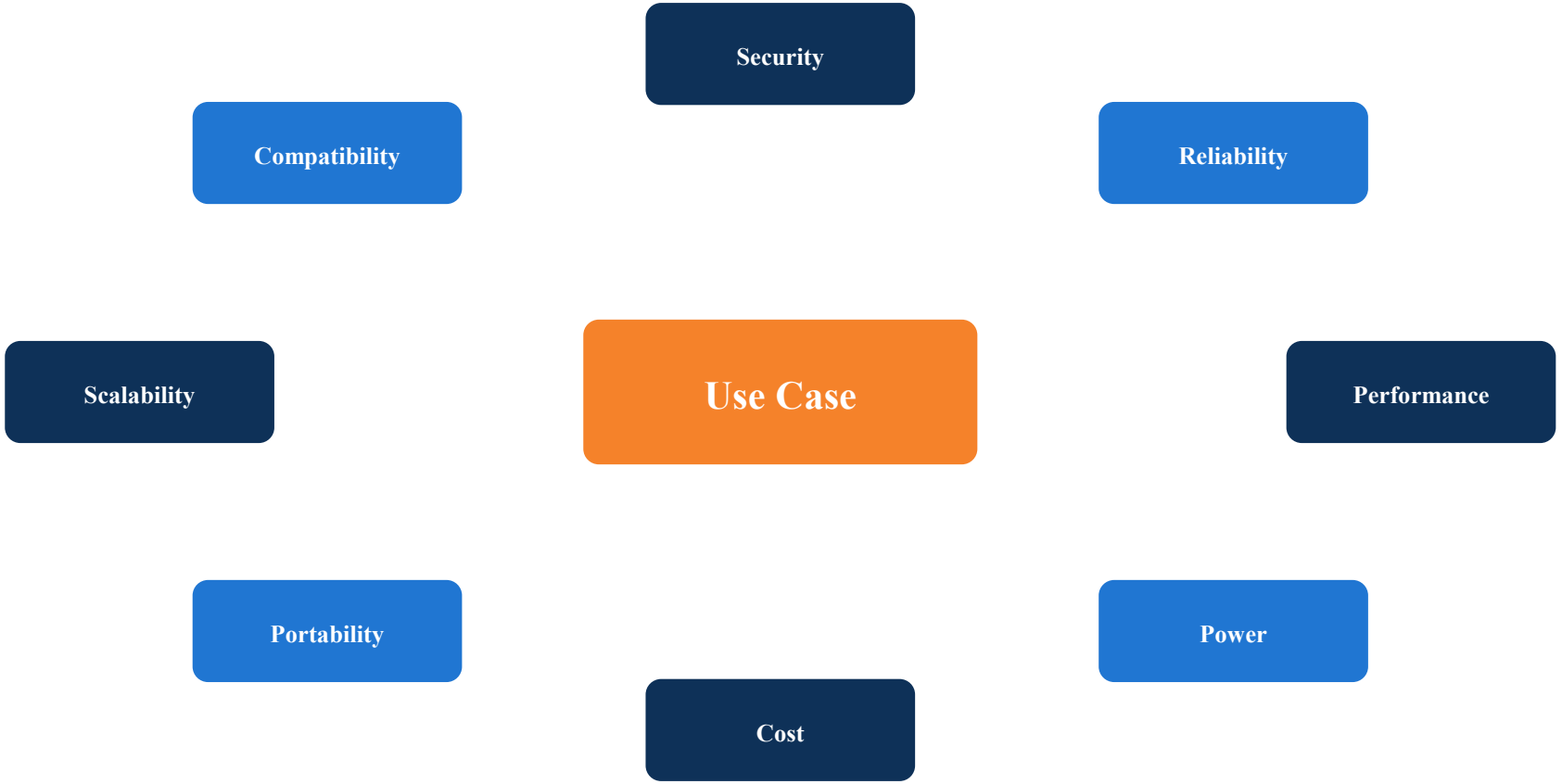
→ **Preview**

IaaS • PaaS • SaaS

Detailed in Week 4.

Choosing a Platform Is About Trade-offs

There is no universally “best” computer.



Decision principle
Start with the problem, then select the hardware, software, and services that fit.

Why Understand Computer Platforms?

Platform knowledge supports better technical decisions.

✓ Choose

right tools and technology

↔ Connect

compatibility and interoperability

↗ Optimize

performance, cost, and energy

+ Build

development and innovation

🔒 Protect

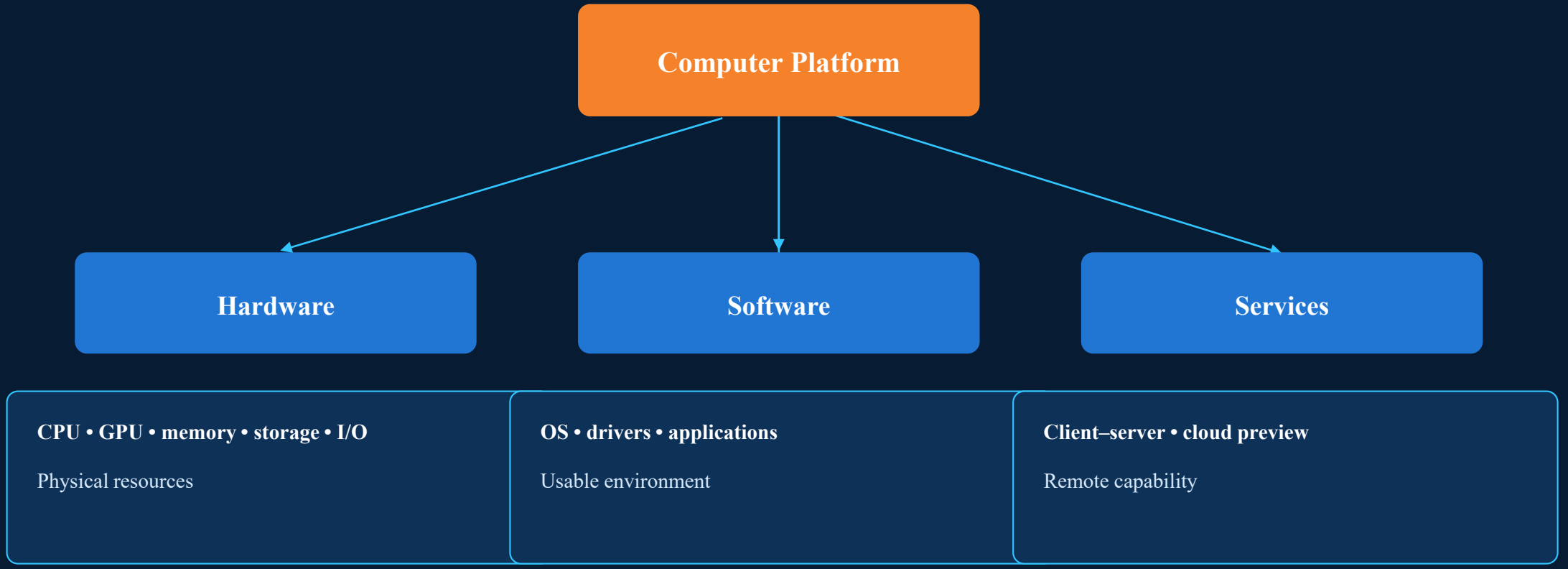
security and reliability

◎ Adapt

new technology trends

Computer Platform in One Picture

Exit question: what makes platforms different?



? **Exit question**

If phones, laptops, servers, and smartwatches are all computers, what makes their platforms different?